

Question

Can we construct a useful evaluation set of experimental protein complexes in which ****every constituent protein chain is below 30% sequence identity to the training data****, and measure exp343's inter-chain contact R-precision and structural accuracy after Helico folding?

The first deliverable is a survival table and an evidence-backed feasibility decision, before running predictors.

Audit contract

Candidates are natural protein-only dimers, with each chain at least 40 residues, total length at most 1,998, and X-ray/EM resolution at most 3.5 Å. FoldBench uses the benchmark's exact ground-truth chain copies; PINDER uses its test split.

Every full and resolved chain representation is searched. A hit requires exact identity $\geq 30\%$ and alignment coverage $\geq 50\%$ of either sequence. A single chain hit rejects the complex. Unsearched arms remain unknown.

Survival

FoldBench: 239 source assemblies -> 184 scoped dimers -> 14 after AFCDB -> 12 after PINDER train -> 4 after AFDB -> **0 after ESM-Atlas**.

PINDER: 1,955 test dimers -> 1,929 scoped dimers -> 40 after AFCDB -> 34 after PINDER train -> 8 after AFDB -> **1 after ESM-Atlas**.

The last PINDER target is 6XNR, a 290-residue homodimer. The two ProteinMPNN arms remain unsearched, so one is an upper bound.

Why the earlier FoldBench filter was insufficient

exp225 searched all FoldBench chains, but its MMseqs search used the default score-derived identity estimate. Exact backtraces move some retained alignments across the 30% boundary.

For 8AXI-A versus one retained ESM-Atlas target, exp225 reported 29.7%. The exact backtrace has 200 identities over 620 aligned positions: 32.26%. This audit stores ``nident`` and applies the threshold with integer arithmetic.

Decision

These sources cannot support a useful held-out evaluation of the current checkpoint. FoldBench has zero local-arm survivors; PINDER has at most one before ProteinMPNN, and 6XNR is exposed to Helico's fine-tuning pool.

Do not launch R-precision or structure predictions on this pool. Search newer experimental complexes against the checkpoint-exact indexes, or freeze the desired evaluation set first and retrain every data arm with that reference excluded.

survival.png

Natural protein-only dimers after scope/quality filters and cumulative sequence exclusion against four locally available exp343 training arms. Every constituent chain must avoid a $\geq 30\%$ identity alignment covering $\geq 50\%$ of the shorter sequence. The remaining PINDER target is an upper bound because the ProteinMPNN redesign corpora were not searched.

